

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 2
DIGITAL SIGNAL PROCESSING - SIMULATION OUTPUT

7. Limit Cycle Oscillation in IIR Filter

IIR Filter: $H(z) = 1 / (1 - 1.4z^{-1} + 0.49z^{-2})$ | Word Length: 8 bits | Input removed after 100 samples

1. Objective

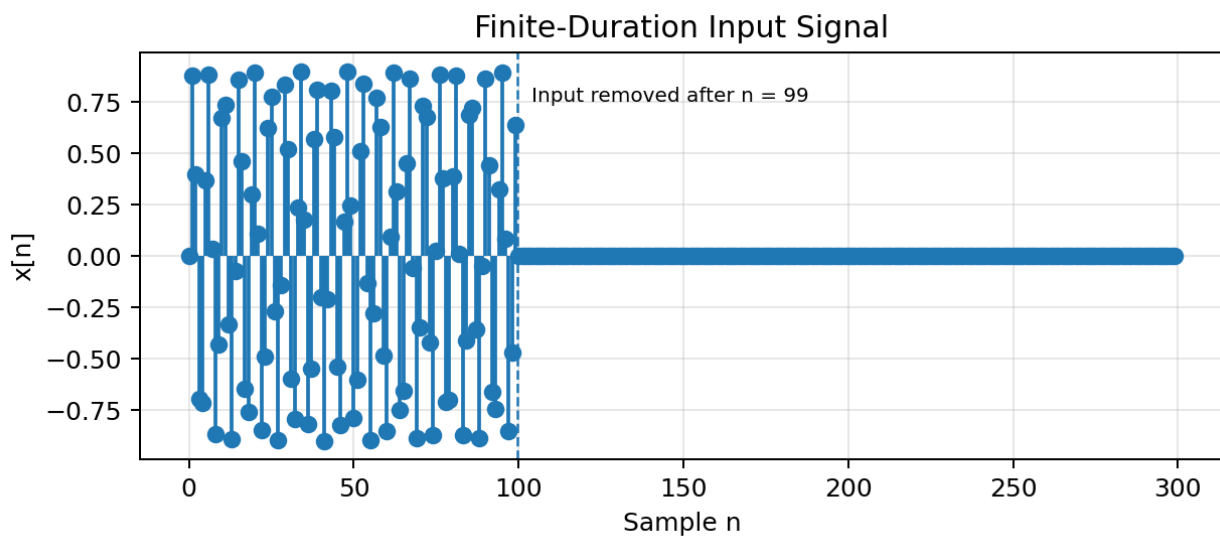
To simulate a second-order fixed-point IIR filter and observe limit-cycle oscillations after the input becomes zero.

2. Difference Equation

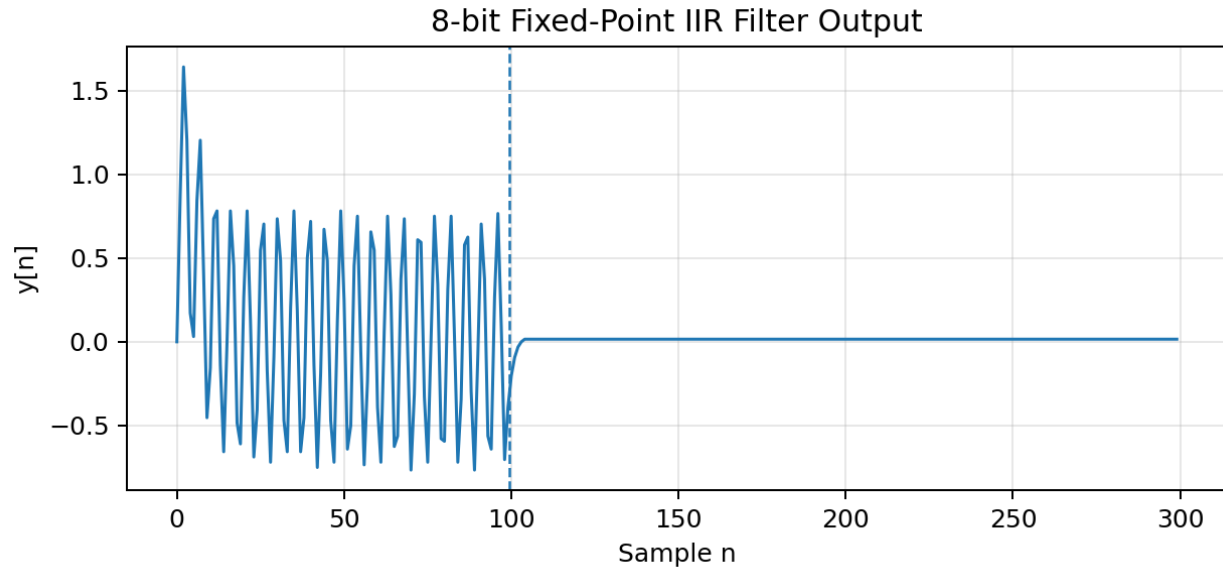
$$y[n] = 1.4y[n-1] - 0.49y[n-2] + x[n]$$

An 8-bit signed fixed-point representation with 6 fractional bits (Q2.6) is used. The feedback arithmetic is quantized after each calculation.

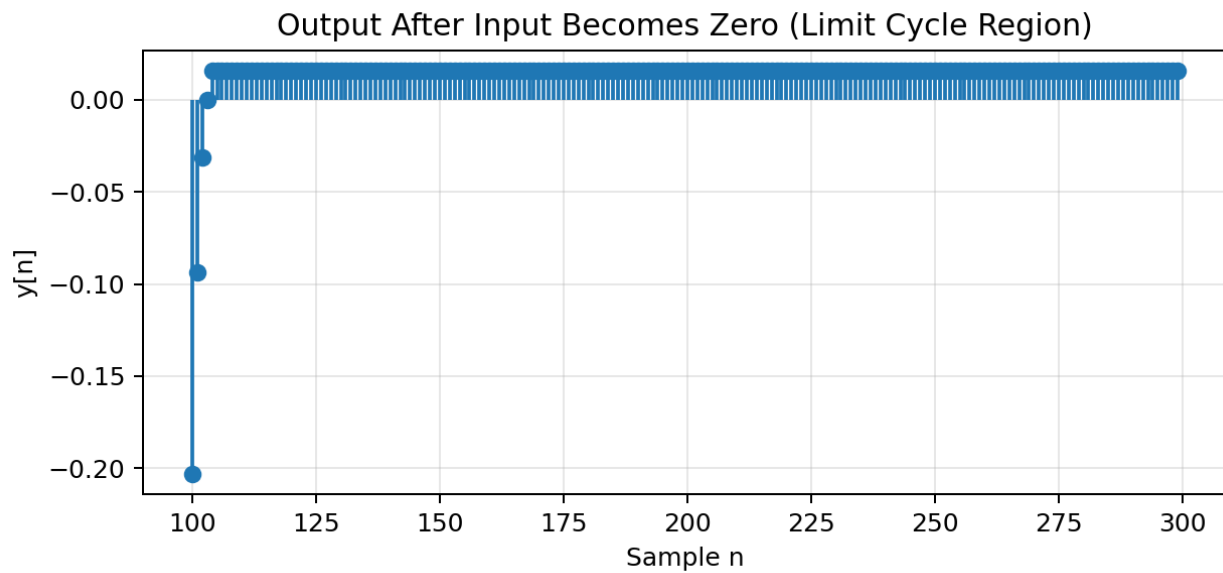
3. Input Signal



4. Filter Output



5. Limit Cycle Observation



6. MATLAB Output / Last Samples

Last 30 output samples:

```
0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156  
0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156 0.0156
```

7. MATLAB Code

```
clc; clear; close all;
N = 300; input_duration = 100; WL = 8; frac = 6;
x = zeros(1,N); n = 0:input_duration-1;
x(1:input_duration) = 0.9*sin(2*pi*0.2133898305*n);
a1 = 1.4; a2 = -0.49;
scale = 2^frac; a1i = round(a1*scale); a2i = round(a2*scale);
qmin = -128; qmax = 127;
xi = round(x*scale); yi = zeros(1,N);
for k = 1:N
    y1 = 0; y2 = 0;
    if k >= 2, y1 = yi(k-1); end
    if k >= 3, y2 = yi(k-2); end
    num = xi(k)*scale + a1i*y1 + a2i*y2;
    yi(k) = max(qmin,min(qmax,round(num/scale)));
end
y = yi/scale;
figure; subplot(2,1,1); plot(0:N-1,x); grid on;
xlabel('Sample n'); ylabel('x[n]'); title('Finite-Duration Input');
subplot(2,1,2); plot(0:N-1,y); grid on;
xlabel('Sample n'); ylabel('y[n]'); title('8-bit Fixed-Point IIR Output');
figure; stem(100:N-1,y(101:end),'filled'); grid on;
xlabel('Sample n'); ylabel('y[n]');
title('Limit Cycle Oscillation After Input Becomes Zero');
disp(y(end-29:end));
```

8. Observation and Analysis

After the input is removed, the output does not immediately settle to exactly zero. Small residual oscillations remain because the 8-bit quantizer introduces finite-word-length errors in the feedback path. These errors are repeatedly fed back by the IIR structure, producing a small sustained periodic or pseudo-periodic output. This is the limit-cycle oscillation.

9. Conclusion

The simulation confirms that finite-word-length effects can produce limit-cycle oscillations in IIR filters. Increasing the word length, applying proper scaling, and using suitable quantization techniques can reduce the oscillation amplitude.